

UNIVERSITY OF GHANA
DEPARTMENT OF COMPUTER SCIENCE

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

ICCTECH

A Web-Based IT Service Management and Helpdesk System

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Live Application: <http://45.79.223.146:8080/index.php>
Source Repository: <https://github.com/Clemzy123/ICCTECH>

1. Purpose, Scope and Product Perspective

This SRS is the final requirements baseline for ICCTECH. It defines the functions, quality attributes, users and acceptance conditions for the 48-hour examination scope. End users submit and monitor requests; analysts triage, prioritise, assign, communicate, update and resolve tickets; administrators manage users, roles, permissions and essential configuration.

Scope

Scope includes authentication and RBAC, unique ticket references, categorisation, priority and assignment, communication and status, resolution, closure and reopen, knowledge, basic assets, user/permission administration, basic operational information, persistent MySQL storage and browser-based cloud access. Wider repository functionality is outside acceptance unless explicitly listed.

Primary users

- **End User** (low privilege): submit and monitor permitted support tickets and use published knowledge.
- **IT Support Analyst** (medium privilege): triage, categorise, prioritise, assign, communicate, update and resolve tickets.
- **System Administrator** (high privilege): manage users, analysts, roles, permissions and essential configuration.
- **IT managers / secondary stakeholders**: receive only granted operational views.

This baseline contains 16 functional requirements and 12 non-functional requirements. Must-Have items define the minimum viable examination workflow; Should-Have and Could-Have items provide supporting capability. Companion verification evidence is maintained in the Testing Report.

2. Overall Description

2.1 Product Perspective

ICCTECH is a centralised browser-based helpdesk application intended to replace fragmented support channels such as verbal reports, telephone calls, messaging applications, email threads and spreadsheets with a traceable ticket lifecycle. The examination deployment operates on a Linode Linux server using Docker Compose, with an Apache/PHP application service and a separate MySQL 8.0 database service.

System boundary

The SRS covers the prioritised ICCTECH examination workflow. The repository contains a broader upstream platform and third-party components; features outside this baseline are not treated as mandatory deliverables unless explicitly listed as requirements below.

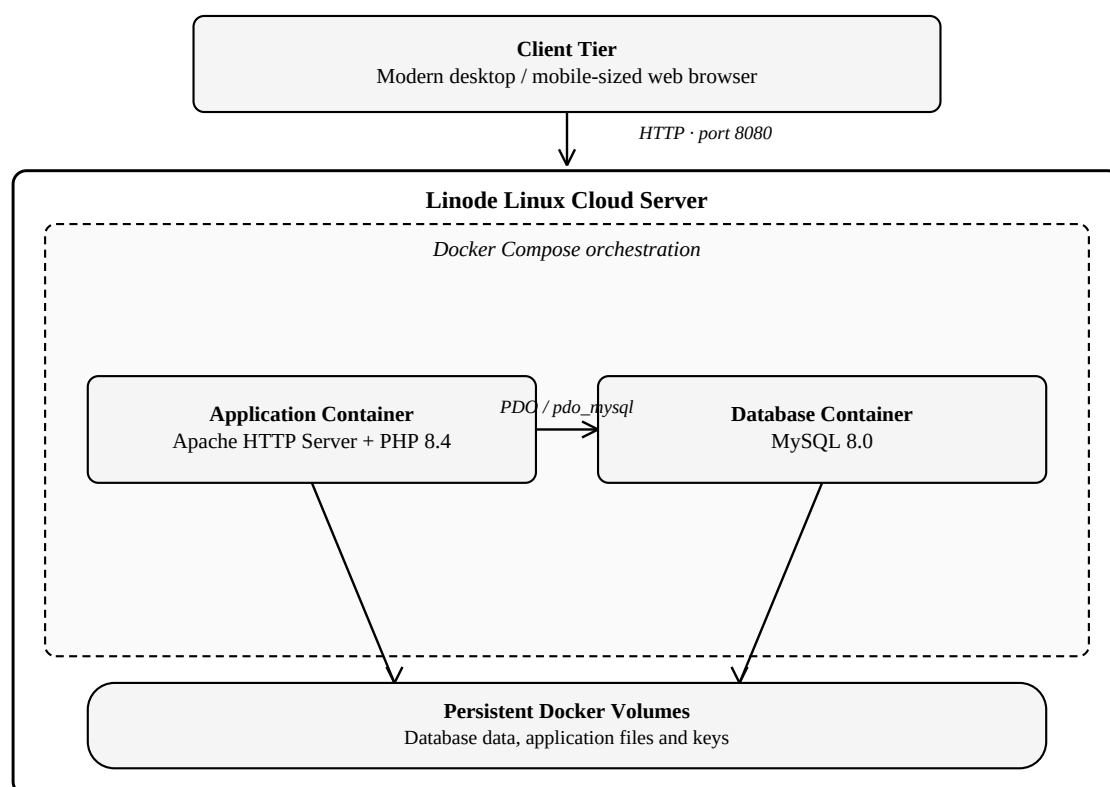


Figure 1. ICCTECH examination deployment architecture, showing the client tier, Linode host, Docker Compose services and persistent volumes (Sections 2.1 and 6.2).

2.2 Major Product Functions

- Authenticate users and enforce role/capability restrictions.
- Allow end users to create and monitor support tickets.
- Generate a unique reference and status for each submitted ticket.
- Allow analysts to triage, categorise, prioritise and assign work.
- Maintain communication and notes within the ticket record.
- Support ticket status transitions, resolution, closure and reopening.
- Provide knowledge-base access for self-service support.
- Maintain relevant IT asset records.
- Allow administrators to manage users, analysts, roles and permissions.

3. Requirements Elicitation and Analysis

Requirements were derived from representative stakeholder scenarios, analysis of typical IT support processes, review of the examination brief, source repository and working system, use-case analysis and validation against the deployed solution.

3.1 Analysed Business Needs

- Every support request requires a persistent, uniquely identifiable record and a visible lifecycle state.
- Users need progress visibility; support teams need ownership, category and priority.
- Communication and investigation history must remain attached to the ticket.
- Privileged functions require role/capability restrictions.
- Operational data must survive a normal service restart.
- The selected workflow must be demonstrable, testable and deployable within the 48-hour examination period.

3.2 Core Ticket Lifecycle

The principal business workflow used to derive and prioritise requirements is Submit Ticket → Triage → Categorise/Prioritise → Assign → Investigate/Communicate → Resolve → Close. A closed ticket may be reopened when additional work is required. The lifecycle, shown in Figure 2, is the primary acceptance path for the examination project.

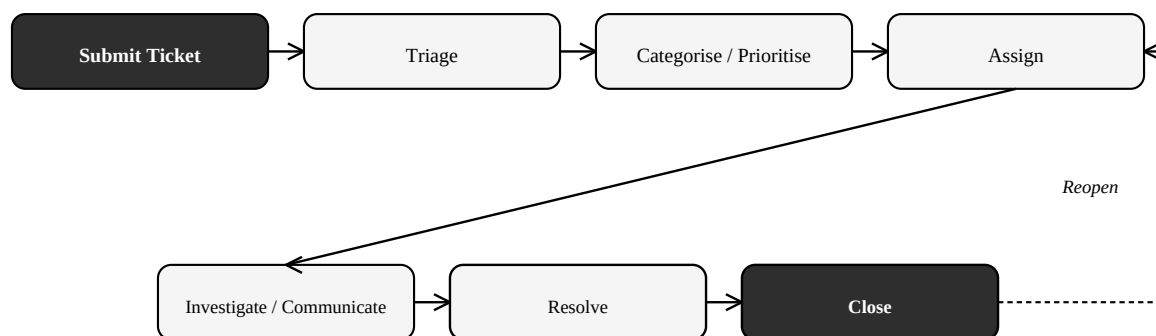


Figure 2. Core ICCTECH ticket lifecycle, including the reopen path (Section 3.2).

4. Functional Requirements

The following functional requirements form the baseline for implementation and verification. “Must” requirements are required for the minimum viable examination workflow; “Should” and “Could” requirements provide supporting capability.

Table 1. Functional requirements baseline

ID	Functional Requirement	Priority	Acceptance Summary
FR-01	The system shall allow authorised users to authenticate using supported credentials.	Must	User presents credentials and, if valid, enters the authorised workspace.
FR-02	The system shall restrict protected functionality according to the authenticated user’s role and capabilities.	Must	Unauthorised functions are not accessible; authorised functions remain available.
FR-03	An end user shall be able to create a support ticket.	Must	A valid ticket submission creates a persistent record.
FR-04	A submitted ticket shall receive a unique reference and an initial status.	Must	The created ticket displays a unique reference and initial lifecycle state.
FR-05	An end user shall be able to view tickets they are authorised to access.	Must	Requester can retrieve permitted tickets without gaining access to unauthorised records.
FR-06	A support analyst shall be able to view incoming tickets.	Must	Authorised analyst can retrieve tickets awaiting support action.
FR-07	A support analyst shall be able to categorise and prioritise a ticket.	Must	Selected category/priority changes are saved and remain visible after reload.
FR-08	A ticket shall be assignable to an appropriate analyst or team where configured.	Must	Assignment is saved and visible as ticket ownership.
FR-09	Authorised users shall be able to add communication or notes to a ticket.	Must	A submitted message/note appears in the ticket record/history.
FR-10	A support analyst shall be able to change a ticket’s status.	Must	Valid status transition is stored and displayed.
FR-11	A support analyst shall be able to record a resolution and close a completed ticket.	Must	Resolution details and closed/resolved state are retained.
FR-12	A closed ticket shall be able to be reopened where further work is required.	Must	Authorised reopen action returns the ticket to an active state and preserves history.
FR-13	An administrator shall be able to manage users, analysts, roles and permissions.	Must	Administrative changes are permitted only to authorised accounts and are retained.
FR-14	Users shall be able to search for and read knowledge articles available to them.	Should	Published/authorised knowledge content can be located and opened.
FR-15	Authorised staff shall be able to create, view or maintain relevant IT asset records.	Should	Permitted asset data can be created/retrieved/updated according to access rights.
FR-16	Authorised administrative users shall be able to view relevant audit or operational information.	Could	Permitted audit/operational views can be accessed without exposing them to unauthorised users.

5. Non-Functional Requirements

The non-functional requirements below define the quality attributes that the delivered system must satisfy, together with the method used to verify each one.

Table 2. Non-functional requirements and verification methods

ID	Category	Requirement	Verification Summary
NFR-01	Usability	Core ticket tasks should be understandable without specialist user training.	Complete representative workflow without specialist instruction.
NFR-02	Performance	Normal pages should respond within approximately three seconds under the expected examination workload, excluding network or hosting delays.	Representative pages meet the target under the defined test conditions.
NFR-03	Security	Protected functions shall require authentication and role/capability checks.	Unauthenticated/unauthorised access attempts are denied.
NFR-04	Security	User input shall be validated before processing or persistence where applicable.	Invalid/unsafe input is rejected or handled without bypassing controls.
NFR-05	Security	User account passwords shall be stored and verified using secure password-hashing mechanisms rather than plaintext storage.	Authentication uses password hashing/verification mechanisms.
NFR-06	Reliability	Tickets, users, assets and other operational records shall be stored persistently in MySQL.	Records remain available after reload and normal service restart.
NFR-07	Availability	The deployed application shall remain accessible for evaluation subject to Linode and network availability.	Live URL responds during verification.
NFR-08	Compatibility	The solution shall operate through modern web browsers in the deployed PHP/MySQL environment.	Core workflow works in the browser used for final evaluation.
NFR-09	Deployability	The application shall be deployable through Docker Compose to a Linux cloud server.	Application and database services start and operate in the target environment.
NFR-10	Maintainability	Project configuration, architecture and key implementation decisions shall be documented.	Submission contains sufficient technical documentation to understand the deployed configuration.
NFR-11	Data Persistence	Operational data, attachments and application keys shall remain available following normal container or service restart through persistent storage.	Controlled service restart does not remove persisted data/files.
NFR-12	Responsiveness	Core pages should remain usable on desktop and mobile-sized screens.	Key pages remain readable and operable at representative desktop/mobile viewport sizes.

6. External Interface Requirements

6.1 User Interface

- The system shall be accessible through a standard web browser.
- Login interfaces shall request appropriate credentials and return clear success/failure behaviour.
- End-user interfaces shall provide ticket creation, ticket viewing and knowledge access.
- Analyst interfaces shall expose permitted ticket triage, assignment, communication and resolution actions.
- Administrator interfaces shall expose user, role, permission and configuration functions according to granted access.
- Core pages should remain usable on desktop and mobile-sized screens.

6.2 Software Interfaces

Table 3. Software interface requirements

Interface	Requirement
MySQL 8.0	The application shall store/retrieve persistent operational data using the configured MySQL database service.
PDO / pdo_mysql	PHP shall communicate with MySQL through the configured PDO database-access layer.
Docker Compose	The application and database services shall be orchestrated as separate services for deployment.
Web Server	Apache shall serve the PHP application within the application container.

6.3 Communication Interfaces

Users communicate with the application over HTTP in the examination deployment using the Linode server and port 8080. HTTPS/TLS is a documented production-hardening requirement and technical-debt item rather than a completed core requirement for the current examination deployment.

6.4 Hardware Interfaces

No specialised client hardware is required. Users need a network-capable device with a compatible web browser. The server-side environment requires sufficient CPU, memory and storage to run the Linux host, application container, MySQL container and persistent volumes.

7. Use-Case Specifications

Each use case below expands the interactions implied by the functional requirements and identifies the requirements it verifies.

UC-01 – Authenticate User

Field	Specification
Primary Actor(s)	End User / Analyst / Administrator
Preconditions	User has an active account.
Trigger	User submits valid credentials.
Main Success Flow	1. User opens login page. 2. User enters credentials. 3. System validates credentials. 4. System establishes an authorised session. 5. User is directed to the permitted workspace.
Alternative / Exception Flow	Invalid credentials are rejected; locked/restricted account remains denied.
Postconditions	Authenticated session exists and access is limited by role/capabilities.
Related Requirements	FR-01, FR-02; NFR-03, NFR-05

UC-02 – Submit Support Ticket

Field	Specification
Primary Actor(s)	End User
Preconditions	User is authenticated and has ticket-submission permission.
Trigger	User selects the new-ticket function.
Main Success Flow	1. User enters required ticket information. 2. System validates input. 3. System creates the ticket. 4. System assigns a unique reference and initial status. 5. System confirms submission.
Alternative / Exception Flow	Validation failure prevents invalid submission and prompts correction.
Postconditions	Persistent ticket exists and is visible to authorised users.
Related Requirements	FR-03, FR-04; NFR-04, NFR-06

UC-03 – View and Track Ticket

Field	Specification
Primary Actor(s)	End User / Analyst
Preconditions	User is authenticated and authorised for the requested ticket.
Trigger	User opens a permitted ticket list or ticket reference.
Main Success Flow	1. System checks permission. 2. System retrieves the ticket and related permitted information. 3. System displays status, ownership and communication history.
Alternative / Exception Flow	Unauthorised ticket access is denied.
Postconditions	User sees the current persistent ticket state.
Related Requirements	FR-05, FR-06; FR-02

UC-04 – Triage, Prioritise and Assign Ticket

Field	Specification
Primary Actor(s)	Analyst / Administrator
Preconditions	Ticket exists; actor has ticket-management capability.
Trigger	Actor opens an incoming ticket.
Main Success Flow	1. Review request. 2. Select category. 3. Set priority. 4. Assign analyst/team. 5. Save changes. 6. System retains updated ownership and triage data.
Alternative / Exception Flow	Invalid or unauthorised changes are rejected.
Postconditions	Ticket has a defined category, priority and owner where applicable.
Related Requirements	FR-07, FR-08

UC-05 – Communicate and Update Ticket

Field	Specification
Primary Actor(s)	End User / Analyst / Administrator
Preconditions	Ticket exists and actor has access.
Trigger	Actor adds a message/note or analyst changes status.
Main Success Flow	1. Actor opens ticket. 2. Actor enters permitted communication/update. 3. System validates and stores it. 4. System displays updated history/status.
Alternative / Exception Flow	Unauthorised notes/status changes are denied.
Postconditions	Communication/status update is retained in ticket history.
Related Requirements	FR-09, FR-10

UC-06 – Resolve, Close and Reopen Ticket

Field	Specification
Primary Actor(s)	Analyst / Administrator
Preconditions	Ticket exists; actor can resolve/close/reopen.
Trigger	Support work is complete or further work becomes necessary.
Main Success Flow	1. Analyst records resolution. 2. System stores resolution. 3. Ticket is marked resolved/closed. 4. If further work is required, authorised actor reopens the ticket. 5. System preserves previous history.
Alternative / Exception Flow	Invalid status transition or unauthorised action is denied.
Postconditions	Ticket reaches the appropriate lifecycle state with history retained.
Related Requirements	FR-11, FR-12

UC-07 – Manage Users, Roles and Permissions

Field	Specification
Primary Actor(s)	Administrator
Preconditions	Administrator is authenticated with required capabilities.
Trigger	Administrator opens user/access administration.
Main Success Flow	1. Search/create/select account. 2. Modify permitted account/role information. 3. System validates privilege. 4. System saves change.
Alternative / Exception Flow	Non-administrator/insufficient capability is denied.
Postconditions	Access configuration is updated and enforced.
Related Requirements	FR-13, FR-02

UC-08 – Search Knowledge Base

Field	Specification
Primary Actor(s)	End User / Analyst / Administrator
Preconditions	User has access to the knowledge function.
Trigger	User searches or browses knowledge content.
Main Success Flow	1. User enters search/browse criteria. 2. System retrieves available articles. 3. User opens an authorised article.
Alternative / Exception Flow	Unavailable/unpublished content is not exposed.
Postconditions	User reads permitted troubleshooting information.
Related Requirements	FR-14

UC-09 – Maintain Asset Record

Field	Specification
Primary Actor(s)	Authorised Analyst / Administrator
Preconditions	Actor has asset-management permission.
Trigger	Actor opens asset function.
Main Success Flow	1. Search/create/select asset. 2. Enter or update permitted information. 3. System validates and stores changes. 4. Updated asset record is displayed.
Alternative / Exception Flow	Unauthorised modification is denied.
Postconditions	Persistent asset information is available to permitted users.
Related Requirements	FR-15

8. Data, Prioritisation and Acceptance Criteria

8.1 Data Integrity and Persistence

Tickets, users, analysts, statuses, priorities, communication/notes, audit/system logs, assets, knowledge and RBAC data must retain referential and access-control integrity. Unique ticket references must remain unique; permitted state, priority, assignment and history updates must persist; MySQL-backed operational records and persistent application data must survive a normal restart.

8.2 MoSCoW Baseline

Must: FR-01 to FR-13. **Should:** FR-14 knowledge and FR-15 assets. **Could:** FR-16 audit/operational information. **Out of scope:** native mobile apps, WhatsApp/SMS, full email-to-ticket, production external identity integration, advanced AI/predictive analytics, large-scale performance testing, enterprise disaster-recovery automation and non-essential integrations.

8.3 Acceptance Criteria

- Valid users authenticate; invalid/unauthorised access is denied.
- Ticket submission creates a persistent unique-reference record visible to the authorised requester.
- Category, priority, ownership, communication and status changes persist.
- Resolution/closure is retained and eligible tickets can reopen without losing history.
- Authorised administrators can manage access; ordinary accounts cannot perform protected administrative actions.
- Knowledge and permitted asset functions are available to authorised users.
- Operational data survives normal restart and the live Linode deployment is reachable.
- Representative normal pages meet the approximately three-second target and core pages remain usable at desktop/mobile viewport sizes.

8.4 Change Control

After baseline, each proposed change identifies affected FR/NFRs, value, effort, risk and schedule impact; accepted changes are re-prioritised and require corresponding SRS, traceability and test updates. Features present in the wider repository are not silently added to scope.

9. Requirements Traceability and Final Baseline

The matrix below is the authoritative mapping from each FR/NFR to its stakeholder or quality concern and verification evidence. The baseline contains 16 functional and 12 non-functional requirements. Traceability is maintained by keeping requirement identifiers stable across the SRS, implementation discussion and Testing Report. A change to a baselined requirement should therefore trigger review of its related design, implementation and test evidence.

Table 4. Requirements traceability matrix

Requirement	Source / Concern	Priority / Type	Verification Evidence
FR-01	Administrator / security	Must	TC-01, TC-02
FR-02	Administrator / security	Must	TC-02, TC-10
FR-03	End user	Must	TC-03
FR-04	End user	Must	TC-03
FR-05	End user	Must	TC-03 / workflow evidence
FR-06	Support analyst	Must	TC-04 / workflow evidence
FR-07	Support analyst	Must	TC-04
FR-08	Support analyst	Must	TC-05
FR-09	End user	Must	TC-06
FR-10	Support analyst	Must	TC-07
FR-11	Support analyst	Must	TC-08
FR-12	Support analyst	Must	TC-09
FR-13	Administrator / security	Must	TC-10, TC-11
FR-14	End user	Should	TC-12
FR-15	Asset/support operations	Should	TC-13
FR-16	Administrator / security	Could	Operational/audit evidence
NFR-01	Usability	Required quality attribute	UAT / workflow observation
NFR-02	Performance	Required quality attribute	TC-16
NFR-03	Security	Required quality attribute	TC-02, TC-10
NFR-04	Security	Required quality attribute	Input-validation/security checks
NFR-05	Security	Required quality attribute	Implementation/security inspection
NFR-06	Reliability	Required quality attribute	TC-03, TC-14
NFR-07	Availability	Required quality attribute	TC-15
NFR-08	Compatibility	Required quality attribute	Browser system test
NFR-09	Deployability	Required quality attribute	TC-15
NFR-10	Maintainability	Required quality attribute	Documentation review
NFR-11	Data Persistence	Required quality attribute	TC-14
NFR-12	Responsiveness	Required quality attribute	TC-17